

Basics	
help <cmd>	Displays information about @cmd usage in GDB
run / r	Executes program in GDB
start	Starts executing program and stops at entry point (main)
next / n [N]	Continues execution till next @N source line
print / p <expr>	Evaluates expression and prints result
list / l [N]	Lists the source code. If line given, lists ±10 lines from line @N
quit / q	Exit GDB
TUI	
Ctrl + x, a	Switch between GDB terminal and TUI
Ctrl + l	Refreshes TUI
layout	Select TUI layouts. Layouts: src, reg, asm
layout split	Splits current display screen
info registers	Shows the contents of registers
disassemble	Displays assembly instructions
nexti / ni [N]	Execute N assembly instructions
step / s	Step inside function
finish / fin	Exit out of function
x[FMT] ADDRESS	Displays content of @ADDRESS based on given @FMT. FMT: (N)(format)(size) i.e. x/9xw 0x800000
continue / c	Continue program execution till next breakpoint.
until / u [loc]	Executes code until @loc. Useful when trying to exit loops
Breakpoints	
break <lockspec>	Adds breakpoint to specified location
info breakpoints	Displays information about all current breakpoints.
delete breakpoints [b]	Deletes all given breakpoints. If none given, deletes all breakpoints
watch <expr>	Breakpoint halting execution of program whenever variable is modified.
info watchpoints	Displays information about all watchpoints.
Prints	
display/[FMT] [address/expr]	Whenever program halts, GDB prints the value of the display variables
info display	Shows active displays
undisp [#num]	Deletes the display @num
dprintf <lockspec> <fmt> <args>	Mix of breakpoint and printf. Gives the same effect of adding printf to your code.
info locals	Displays value of local variables in current scope.
info args	Displays value of function argument.
Backtrace	
backtrace or bt	Prints the callstack
bt full	Same as bt but with more details
frame / f [num]	Goes to frame #num
up [count]	Goes #count frames up
down [count]	Goes #count frames down

Misc	
call <expr>	Same as print but also applicable to call void functions.
finish	Continues execution till end of function
ret <expr>	Returns value of expression
skip <function>/<file>	When using step, skips going into @function or functions listed in @file
info skip	Displays details about the specified skips
save breakpoints <filename>	Saves all current breakpoints to @filename
source breakpoints <filename>	Loads breakpoint information from filename

Breakpoints

disable [b]	Disables breakpoint @b. Breakpoint @b would have no effect on program.
enable [b]	Enables breakpoint @b
enable once [b]	Breakpoint @b is disabled after it is hit once
enable count <N> 	Breakpoint @b is disabled after it is hit @N times.
condition <expr>	Breakpoint @b is triggered only when @expr is true.
condition 	Removes condition from breakpoint @b
ignore <count>	Set an ignore count = @N on @b. Breakpoint remains disabled while ignore count > 0.

Setting variables

Set var <var> = <value>	Sets @var in current scope to value <value>
p <var> = <value>	Same as using set var
set \$<reg> = <value>	Sets value of register @reg to <value>
set args <value>	Sets value of function arguments to given list of <value>
set \$<var> = <value>	Sets a convenience variable with value <value>
*array@len	Displays @len content of array.

Reverse Debugging

record [all]	Starts recording program execution. This is required to run reverse debugging commands.
reverse-next / rn	Same as next but program executes backwards.
reverse-step / rs	Same as step but program executes backwards.
reverse-continue / rc	Same as continue but program executes backwards.
record save <file>	Save execution log in @file.
record restore <file>	Loads execution log from @file.
record stop / s	Stop the record/replay target

Checkpoints

checkpoint	Save a snapshot of the debugged program's current execution state.
info checkpoints	Lists saved checkpoints. Contains: Checkpoint ID, Process ID, Code Address and source-line/labels
restart <checkpoint-id>	Restore the program state that was saved as checkpoint @checkpoint-id.
delete checkpoint <checkpoint-id>	Delete the previously-saved checkpoint @checkpoint-id.